

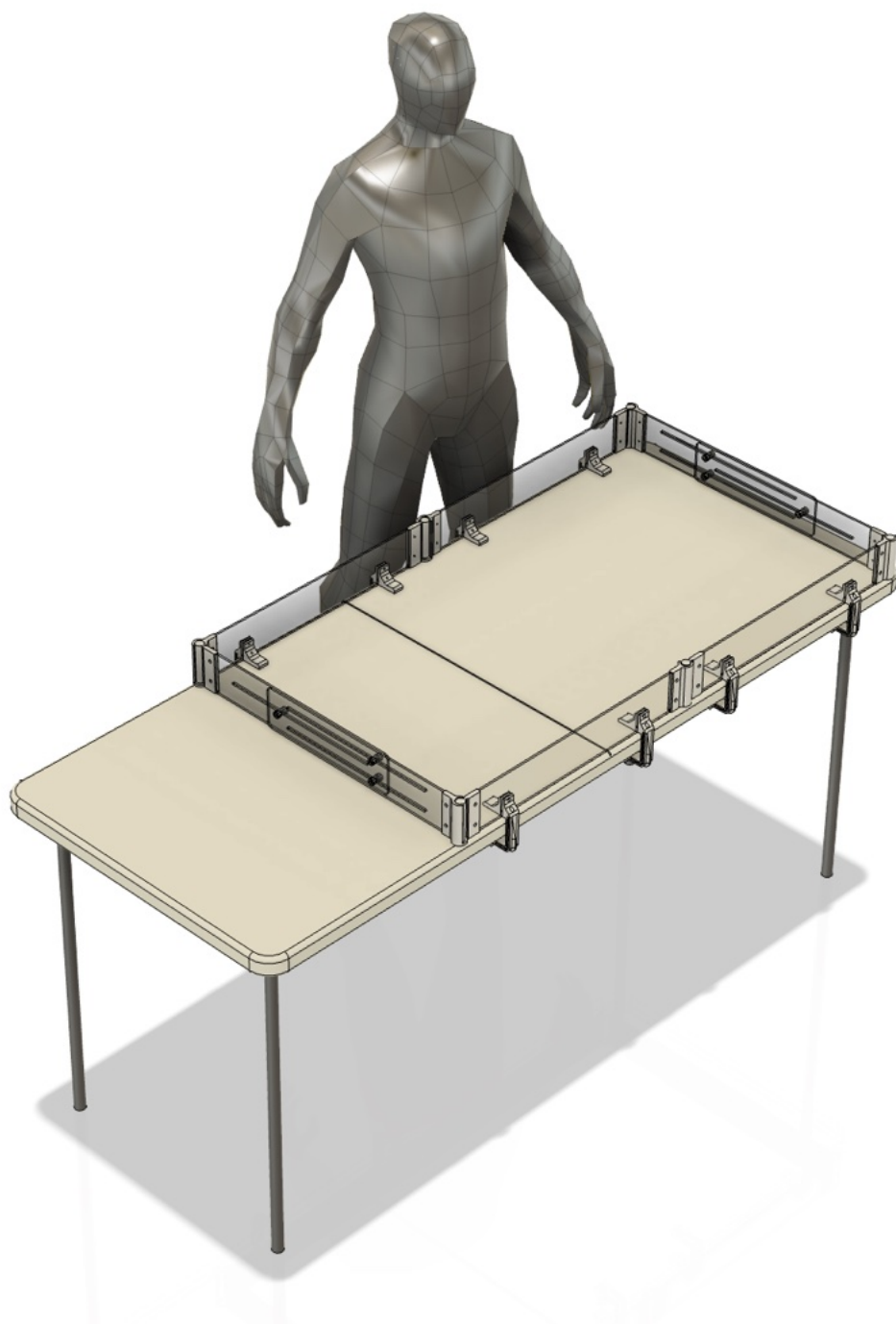


Table Barriers Build Guide

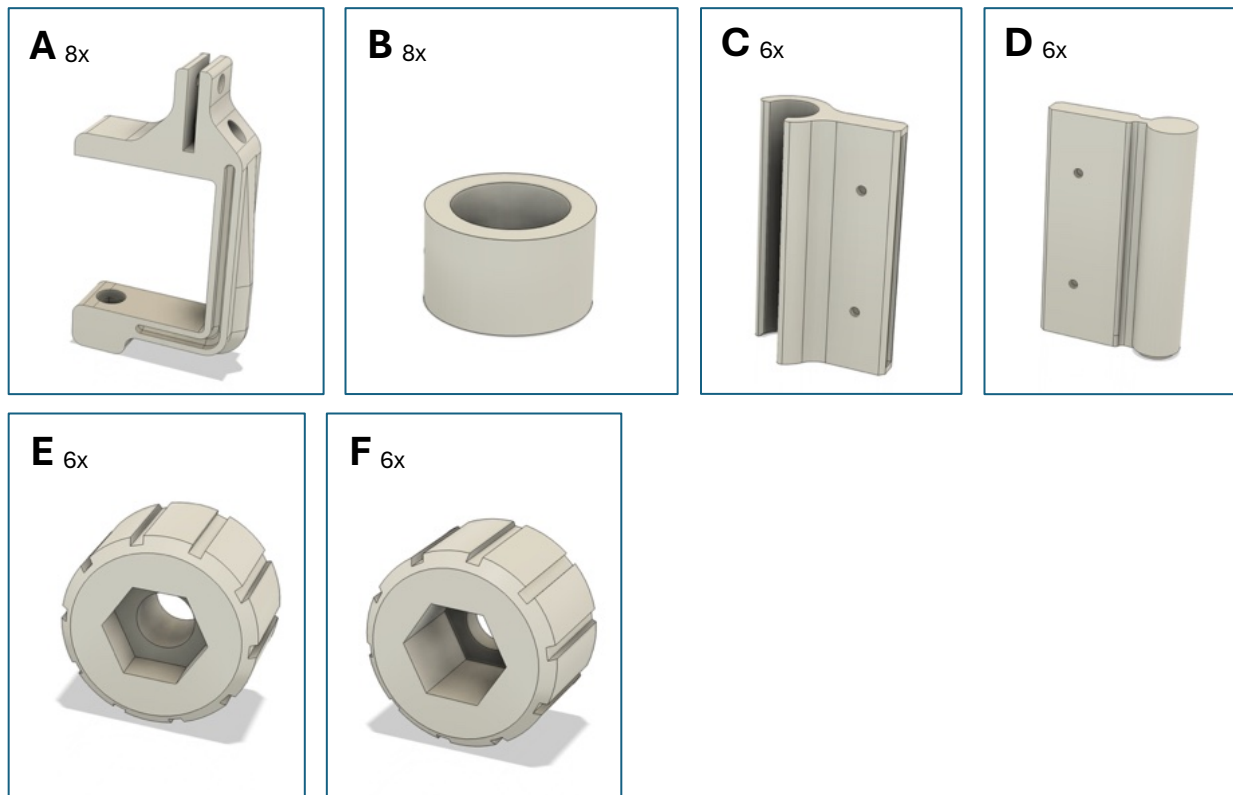
Memphis Zoo

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Parts:

3D Printed Parts:

Manufacturing note: Using high wall counts will ensure that 3D printed parts can stand up to the forces they encounter in regular use of the table barriers. This is particularly important for part A.



Laser Cut Parts:

Manufacturing note: All laser cut parts are designed to be cut from a single sheet of 36" x 24" x 0.177" cast acrylic

G 4x



H 2x

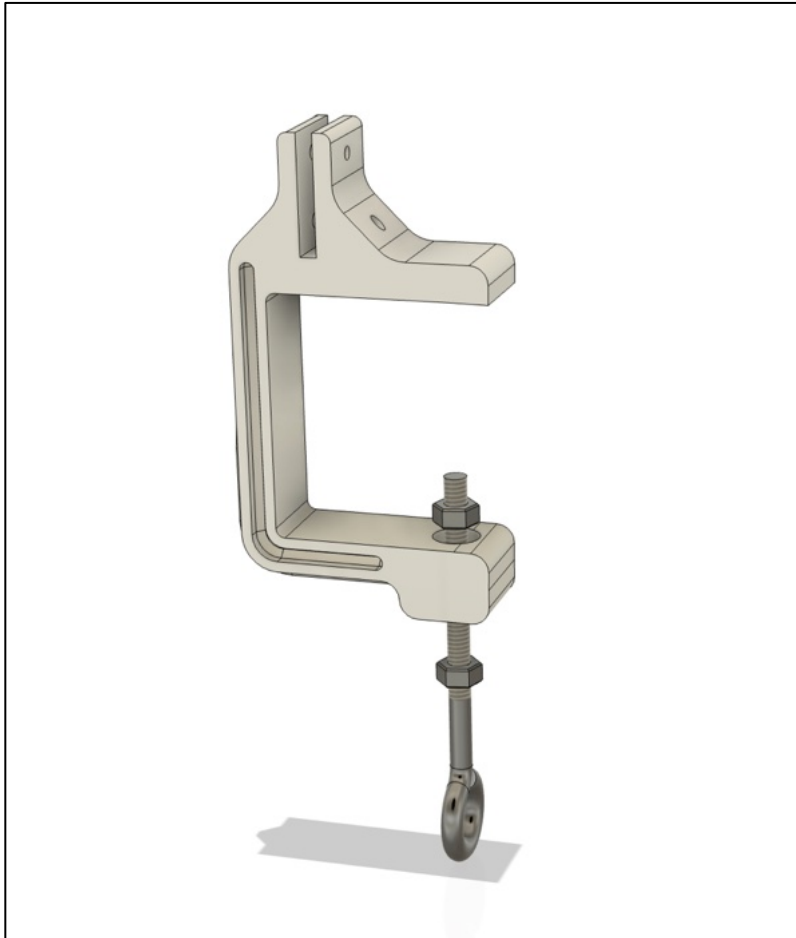


I 2x



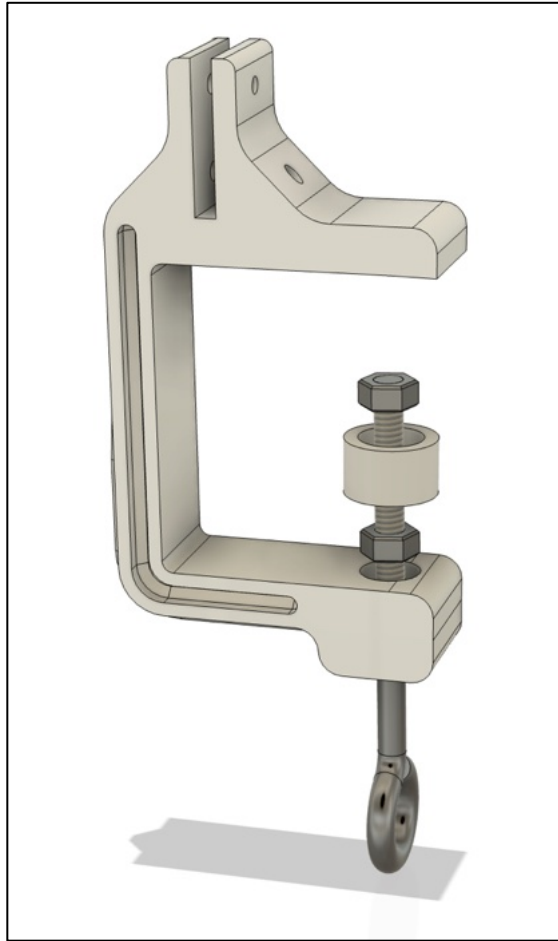
Assembly Guide:

Step 1:



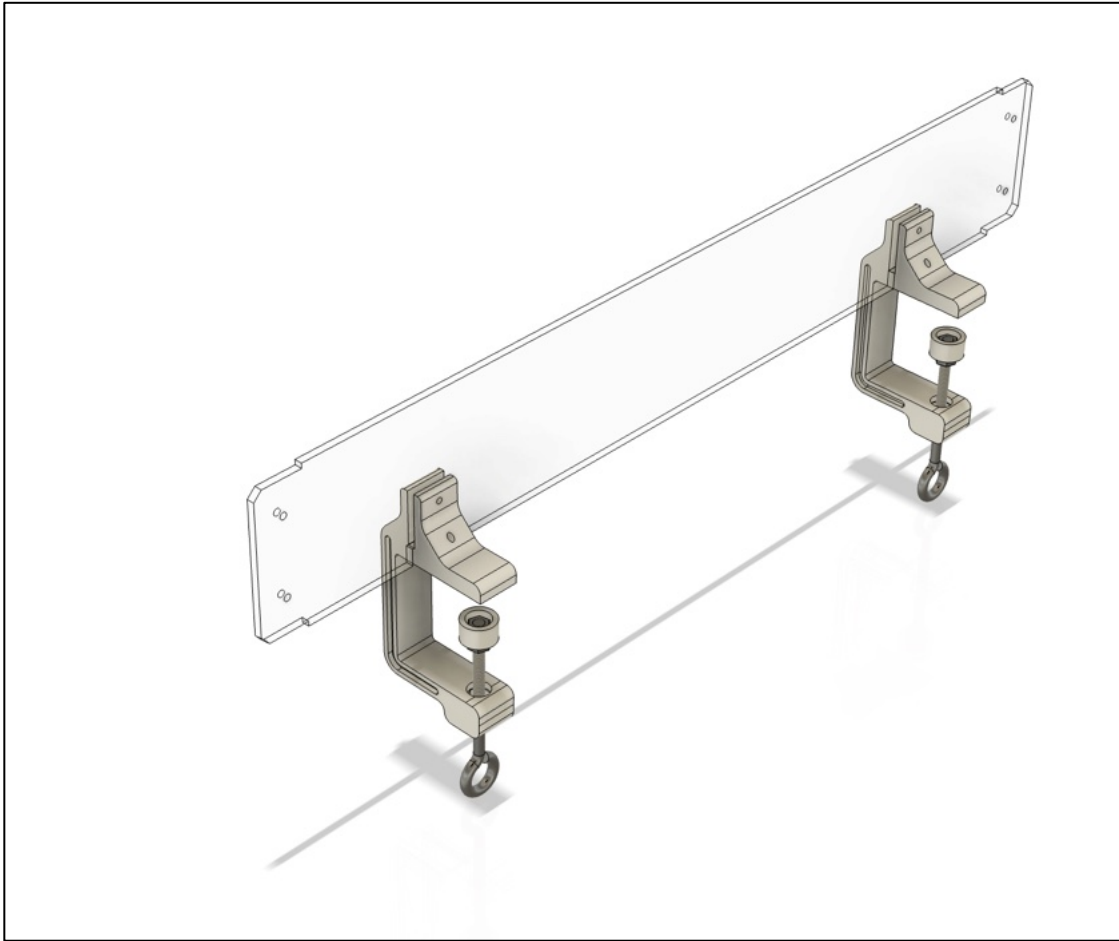
- Seat a 1/4" nut in the recessed hexagonal indentation of part **A** by tightening a second 1/4" nut on a 4" x 1/4" eyebolt against the first below the lowest part of part **A**. After the nut has been seated remove the second nut and screw the eyebolt back into part **A**.

Step 2:



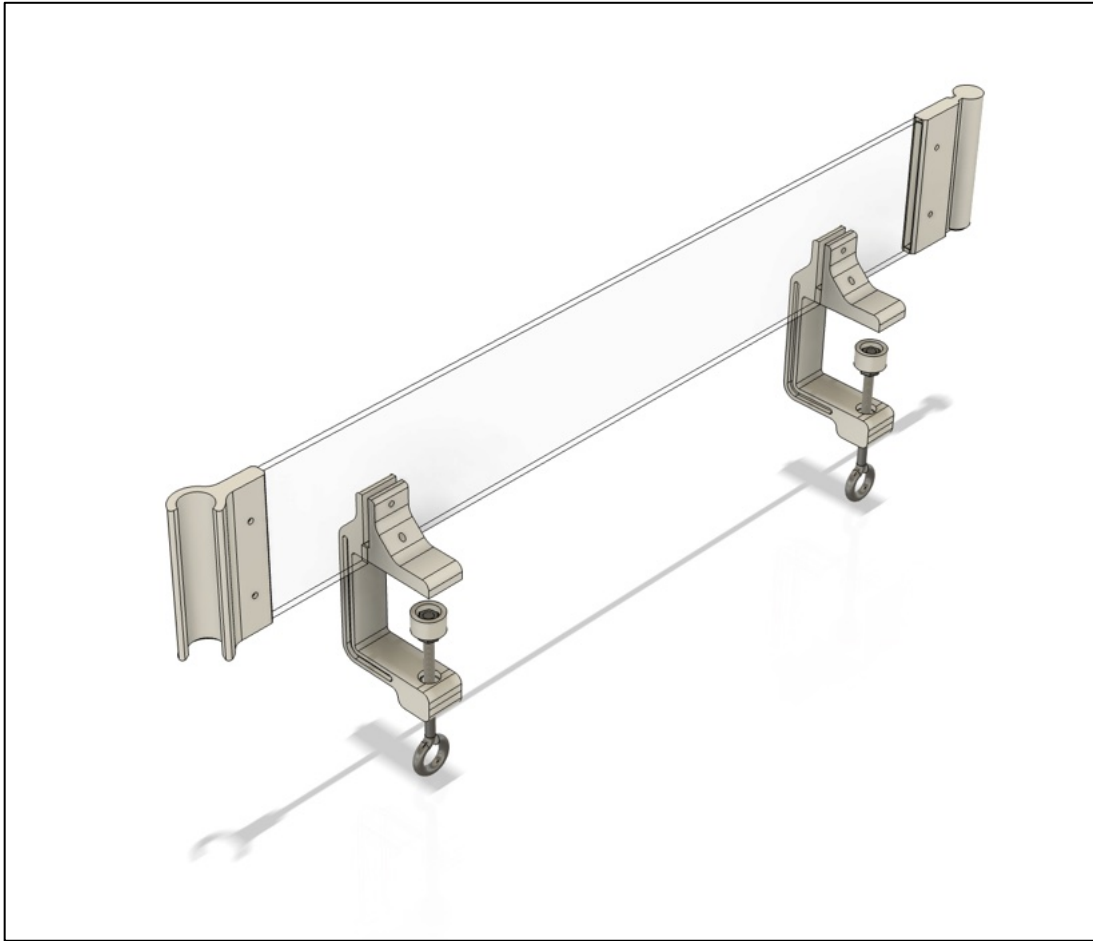
- Screw a 1/4" nylon lock nut on to the eyebolt, insert part **B**, and add another nylon lock nut above part B screwed as far up on the lock nut as it will engage and lock with the threads. Tighten the lower lock nut as high as it will go while still allowing part B to easily turn. Repeat steps 1 and 2 for all 8 instances of parts **A** and **B**. This should leave you with 8 subassemblies containing both part **A** and **B**.

Step 3:



- Slot 2 of the 8 of the subassemblies from steps 1 and 2 on to the indentations in all 4 instances of part **G**. Secure both with 2 #8 x ½" machine screws each. You should now have 4 subassemblies containing parts **A**, **B**, and **G**.

Step 4:



- Slot part **C** on the left end and part **D** onto the right end of each of the 4 subassemblies created in step 3. Secure both with 2 #8 x 1/2" machine screws each. You should now have a total of 4 final side assemblies each containing parts **A**_{x2}, **B**_{x2}, **C**, **D**, and **G**.

Step 5:



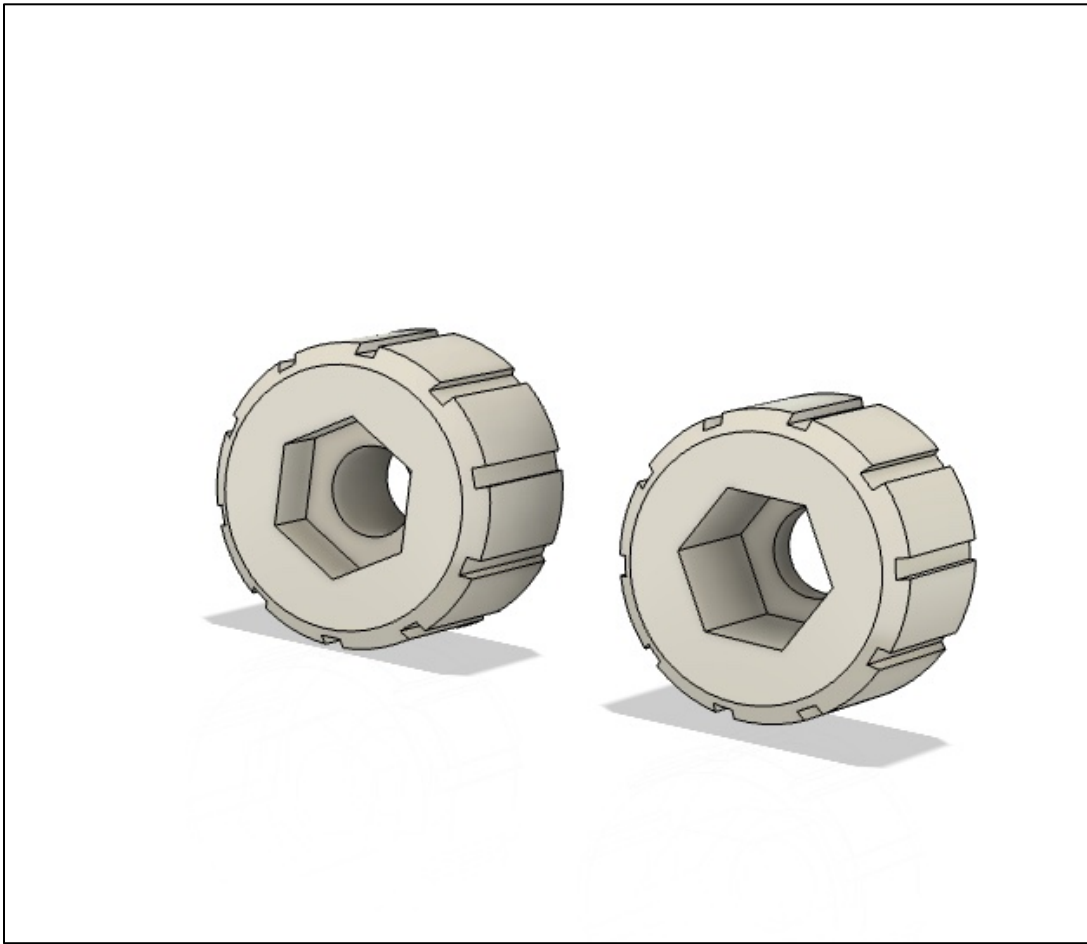
- Slot part **D** onto part **H** and secure it with 2 #8 x ½” machine screws. Repeat this for the other instance of part **H**. This should leave you with two subassemblies containing part **D** and part **H**.

Step 6:



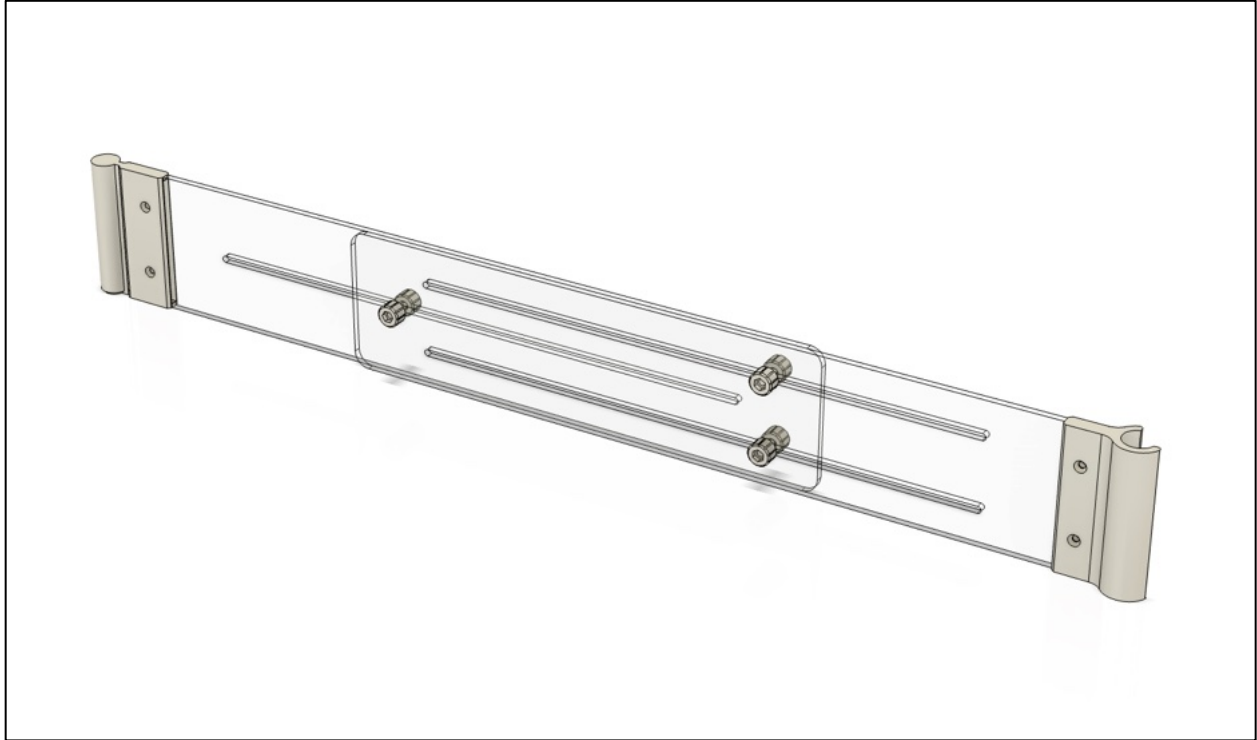
- Slot part **C** onto part **I** and secure it with 2 #8 x ½” machine screws. Repeat this for the other instance of part **I**. This should leave you with two subassemblies containing part **C** and part **I**.

Step 7:



- Press an M4 nut into part **F** (deeper indentation) and a 25mm long M4 bolt into part **E** (shallower indentation). Tighten the parts together to seat the nut and bolt into the recesses. Repeat this step for all 6 instances of parts **F** and **E**.

Step 9:



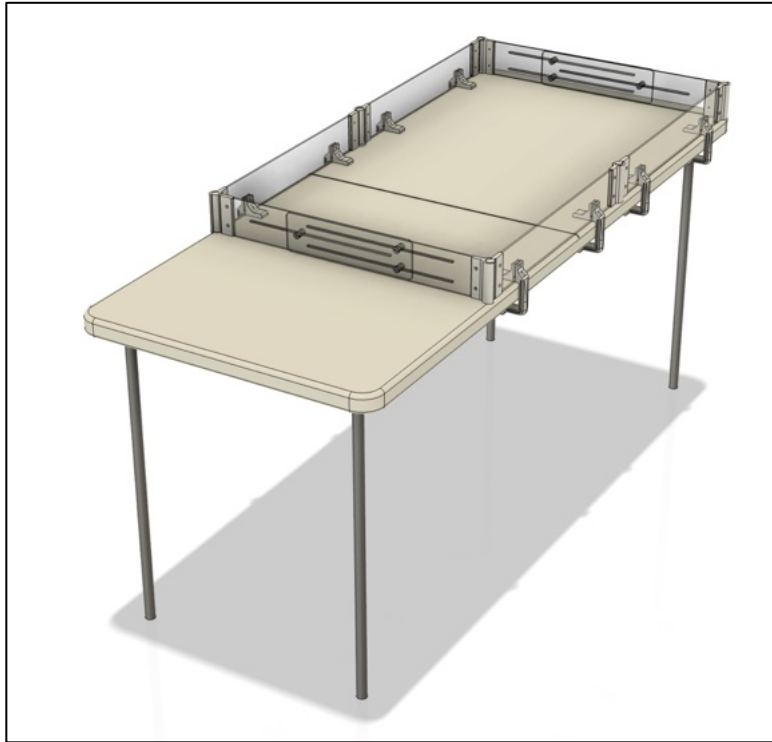
- Use parts **F** and **E** with the seated nut and bolt respectively to connect the subassemblies created in steps 5 and 6. Repeat this for both instances of these assemblies leaving you with two final end assemblies containing parts **C**, **D**, **E_{x3}**, **F_{x3}**, **H**, and **I**.

Step 10:



- Assemble both sides of the barrier by combining all 4 of the side assemblies made in step 4. Attach them to the table by tightening the eye bolts threaded through the nuts in parts **A_{x8}**. If the barriers are not secure enough on the table paint the contact surfaces of parts **A_{x8}** and **B_{x8}** with Plasti Dip using painters tape to mask off non-contact surfaces.

Step 11:



- Drop in both final end assemblies created in step 9 using the 3 **F_{x6}** and **E_{x6}** part pairs on each end assembly to adjust and secure the end lengths to match the width of the table.

If you made it this far, congratulations on your new table barriers! I would like to thank the Memphis Zoo interpretation department for supporting the development of this project.

If you have any alterations you would like to suggest or want to share pictures of the table barriers in use contact me at:

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If you have any enrichment you think would be at home on the Open Source Enrichment website, reach out and I will add them!